

SECTION-I

This section contains **10 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct. **(+3, -1)**

1. If the expression $\begin{vmatrix} x^2 + x + 3 & 1 & 4 \\ 2x^4 + x^3 + 2x + 1 & 2 & 3 \\ x^2 + x & 1 & 1 \end{vmatrix}$ is equal to $ax^4 + bx^3 + cx^2 + dx + e$, then the value of e is equal to
(A) zero (B) 1 (C) 2 (D) none of these
2. For $A = a^2 + b^2 + c^2$, $B = ab + bc + ca$, $(a^3 + b^3 + c^3 - 3abc)^2$ is equal to
(A) $\begin{vmatrix} B & A & B \\ B & B & A \\ A & B & B \end{vmatrix}$ (B) $\begin{vmatrix} A & B & B \\ B & B & A \\ B & A & B \end{vmatrix}$ (C) $\begin{vmatrix} B & B & A \\ B & A & B \\ A & B & B \end{vmatrix}$ (D) none of these
3. If maximum and minimum values of the determinant $\begin{vmatrix} 1 + \sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1 + \cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1 + \sin 2x \end{vmatrix}$ are α and β , then
(A) $\alpha + \beta^{99} = 5$
(B) $\alpha^3 - \beta^{17} = 25$
(C) $(\alpha^{2n} - \beta^{2n})$ is always an even integer for $n \in \mathbb{N}$
(D) a triangle can be constructed having its sides as α , β and $\alpha - \beta$
4. If $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, then the matrix given by $B = I + A + A^2 + \dots + A^k$, is
(A) $\begin{bmatrix} k & 1 \\ 1 & k \end{bmatrix}$ (B) $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 \\ 0 & k+1 \end{bmatrix}$ (D) none of these
5. If A is a $n \times n$ matrix such that $a_{ij} = \sin^{-1} \sin(i - j) \forall i, j$ then which of the following statement is not true
(A) if n is even, then A is an invertible matrix
(B) if n is odd, then A is an invertible matrix
(C) for all values of n, A is not invertible matrix
(D) none of these

6. If ω is the complex cube root of unity, then $\begin{bmatrix} \omega^2 & -\omega^2 \\ \omega & \omega \end{bmatrix}^{-1} \begin{bmatrix} \omega & \omega^2 \\ -\omega & \omega^2 \end{bmatrix}^{-1}$ is equal to
- (A) $\begin{bmatrix} \omega & \omega^2 \\ \omega & \omega \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ (D) none of these
7. If x, y, z are in G.P ($x, y, z > 0$), then the determinant $\begin{vmatrix} xp+y & x & y \\ py+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix}$ is equal to
- (A) 1 (B) 0 (C) a multiple of p (D) None of these
8. If α, β, γ are the roots of the equation $(x^3 + x^2 + x + 1) = 0$ then $\begin{vmatrix} 1+\alpha^2 & 1 & 1 \\ 1 & 1+\beta^2 & 1 \\ 1 & 1 & 1+\gamma^2 \end{vmatrix}$ is equal to
- (A) 1 (B) 0 (C) 2 (D) none of these
9. If $\begin{vmatrix} \sum_{k=0}^{n-2} 1 & n(n-1) & n^2 \\ \sum_{k=1}^n 1 & (n+1)(n-1) & n(n+1) \\ \sum_{k=1}^{n-1} 1 & n^2+1 & n^2 \end{vmatrix} = 72$, then n is equal to
- (A) 6 (B) 9 (C) 8 (D) none of these
10. Let A and B are two square matrices of order n . If $A^\theta A = I$ and $B^\theta B = I$ (where A^θ is the conjugate transpose of A) then $(AB)^\theta$ is equal to
- (A) AB (B) $(AB)^{-1}$ (C) $(BA)^{-1}$ (D) None of these

SECTION-II

This section contains **7 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct. **(+4, -1)**

11. If $a_i, i = 1, 2, \dots, 9$ are perfect odd squares, then $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is always a multiple of
- (A) 4 (B) 7 (C) 16 (D) 5
12. If a, b, c are even natural numbers, then $\Delta = \begin{vmatrix} a-1 & a & a+1 \\ b-1 & b & b+1 \\ c-1 & c & c+1 \end{vmatrix}$ is a multiple of
- (A) 4 (B) 6 (C) 9 (D) None of these
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13. If A and B are invertible matrices both of order n , then
 (a) $\text{adj. } A = |A| A^{-1}$ (b) $(A + B)^{-1} = A^{-1} + B^{-1}$
 (c) $(AB)^{-1} = B^{-1}A^{-1}$ (d) $|A^{-1}| = |A|^{-1}$
14. If $A = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$ is n^{th} root of I_2 then choose the correct statements
 (A) if n is odd, $a = 1, b = 0$ (B) if n is odd, $a = -1, b = 0$
 (C) if n is even, $a = 1, b = 0$ (D) if n is even, $a = -1, b = 0$
15. If $A = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$ is n^{th} root of I_2 then choose the correct statements
 (A) if n is odd, $a = 1, b = 0$ (B) if n is odd, $a = -1, b = 0$
 (C) if n is even, $a = 1, b = 0$ (D) if n is even, $a = -1, b = 0$
16. Let $A = \begin{bmatrix} x^2 & 6 & 8 \\ 3 & y^2 & 9 \\ 4 & 5 & z^2 \end{bmatrix}$, $B = \begin{bmatrix} 2x & 3 & 5 \\ 2 & 2y & 6 \\ 1 & 4 & 2z-3 \end{bmatrix}$. If $\text{trace } A = \text{trace } B$ then $x + y + z$ is equal to
 (A) 0 (B) 6 (C) 5 (D) none of these
17. Let A be a skew-symmetric matrix of odd order, then $\det(A)$ cannot be
 (A) 0 (B) ± 1 (C) ± 2 (D) ± 3

SECTION-III (Paragraph Type)

This section contains **2 groups of questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct. **(+3, -1)**

- I. If A be a square matrix of order n . Then
 $A(\text{adj}A) = |A| I_n = (\text{adj}A)A$
- II. If A and B are non-singular square matrices of the same order, then
 $\text{adj}AB = (\text{adj}B)(\text{adj}A)$
- III. If A be a non-singular square matrix of order n . Then
 $|\text{adj}A| = |A|^{n-1}$.
- IV. If A is a non-singular square matrix, then
 $\text{adj}(\text{adj}A) = |A|^{n-2}A$
18. If A is a square matrix, then $\text{adj}A^T - (\text{adj}A)^T$ is equal to
 (A) $2|A|$ (B) $2|A|I$ (C) null matrix (D) unit matrix
19. Let A be a non-singular square matrix. Then $\det(\text{adjadj}A)$ is equal to
 (A) $|A|^n$ (B) $|A|^{(n-1)^2}$ (C) $|A|^{n-2}$ (D) none of these
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SECTION-IV (Integer Answer Type)

This section contains **2** questions. The answer to each of the questions is a **single digit integer**, ranging from 0 to 9. The correct digit below the question number in the ORS is be bubbled.
(+4, 0)

20. In the system of equations

$$ax + by + cz = A$$

$$bx + cy + az = B$$

$$cx + ay + bz = C,$$

A, B, C are the cofactors of a, b, c in the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$.

Then the value of $a^3 + b^3 + c^3 - 3abc + x^3 + y^3 + z^3 - 3xyz$ is

21. A is a $n \times n$ matrix ($n > 2$) $[a_{ij}]$ where $a_{ij} = \cos\left(\frac{(i+j)2\pi}{n}\right)$. Then the determinant A =

(ANSWER KEY)

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|---------|---------|-----------|-----------|-----------|
| 1. A | 2. A | 3. C | 4. C | 5. B |
| 6. C | 7. B | 8. B | 9. C | 10. B |
| 11. A,C | 12. D | 13. A,C,D | 14. A,C,D | 15. A,C,D |
| 16. D | 17. B,C | 18. C | 19. B | 20. 0. |
| 21. 0 | | | | |

